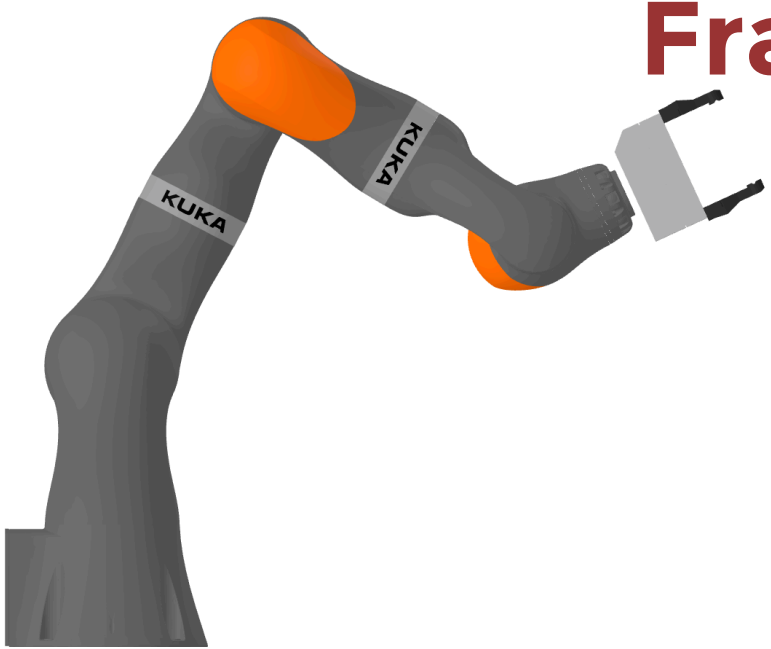


Frames and Transforms

MIT 6.4210/2: Robotic Manipulation

Fall 2026, Lecture 3

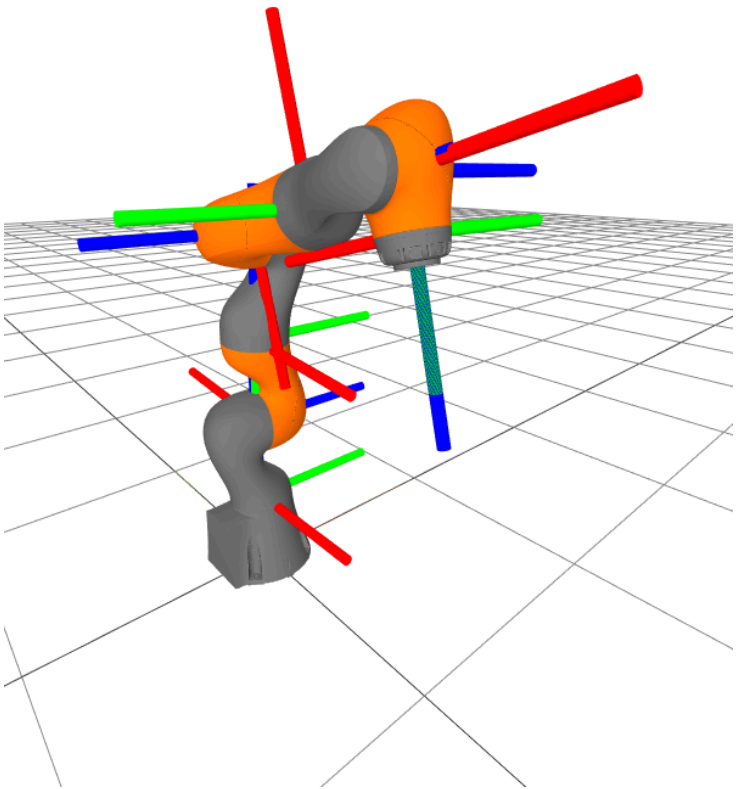
Massachusetts Institute of Technology

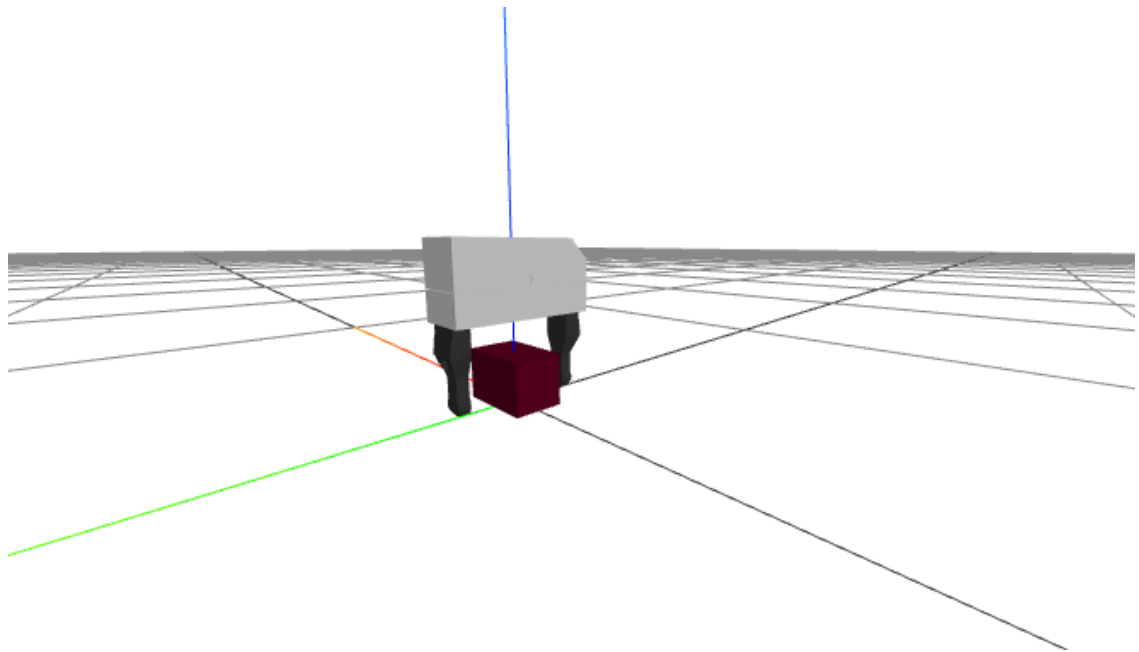
Why learn the basics

- Real robot systems (even "end-to-end" ML) are built on a control and perception stack.
- You need to understand the constraints imposed by the real systems.
- You need to understand what your coding agent produces.

Basic Pick and Place

Step 1: Kinematic Frames / Spatial Algebra

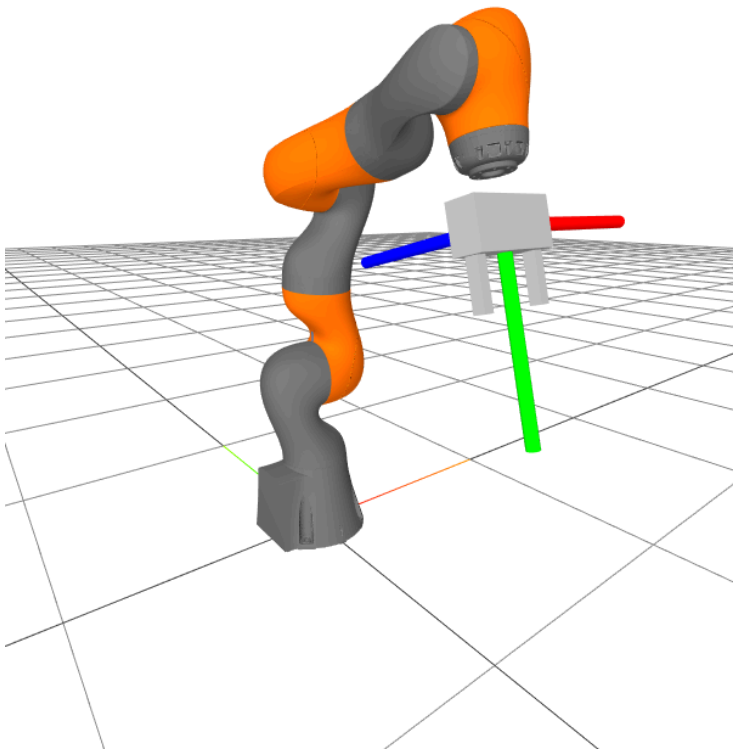


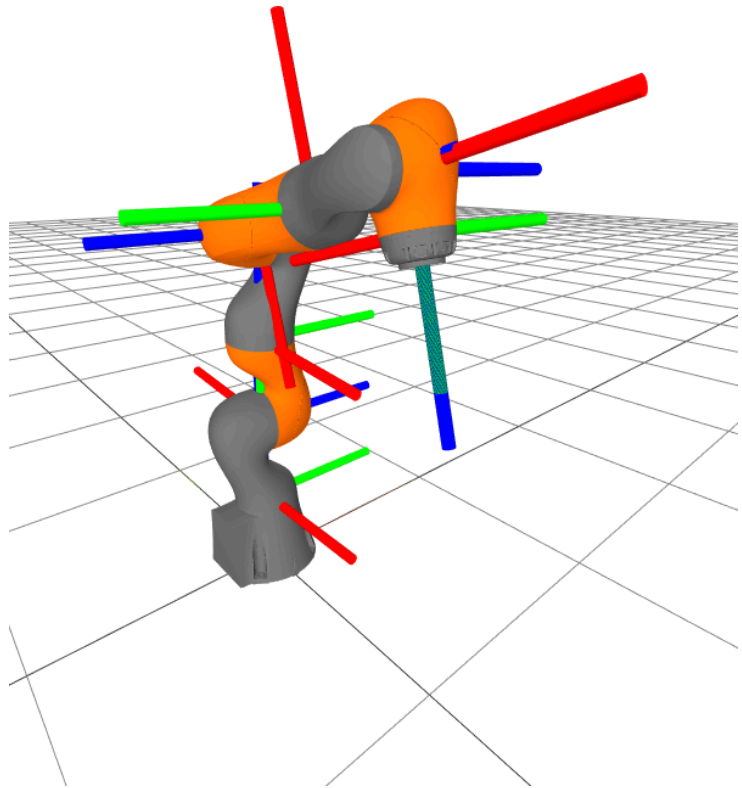


Step 2: Gripper Frame Plan "Sketch"

https://manipulation.mit.edu/data/pick_sketch.html

Step 3: Forward kinematics of the iiwa + WSG





Step 4: Differential Inverse Kinematics

Kinematic Jacobian: $^{W}V^{G} = J^{G}(q)v$

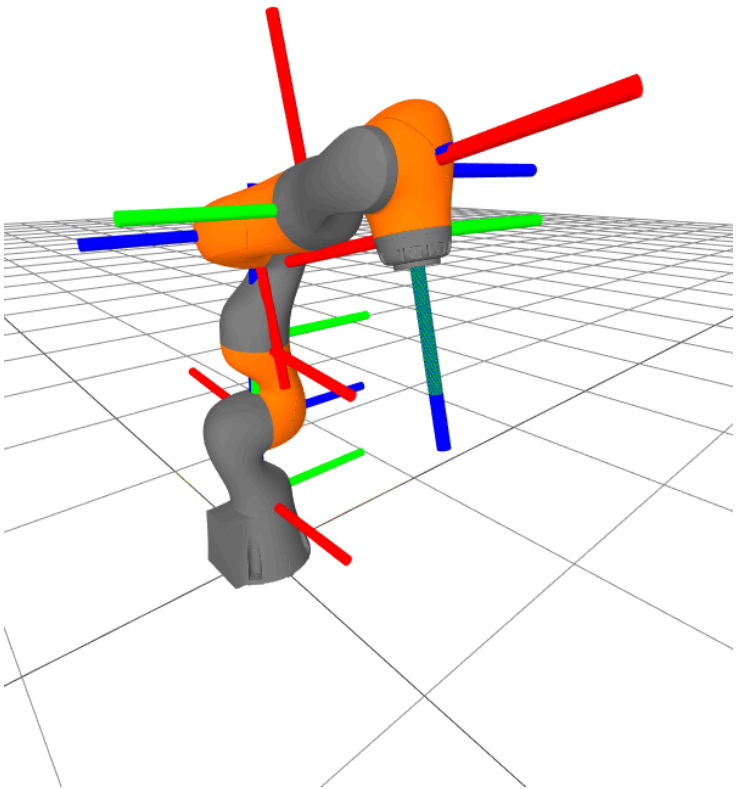
spatial velocity

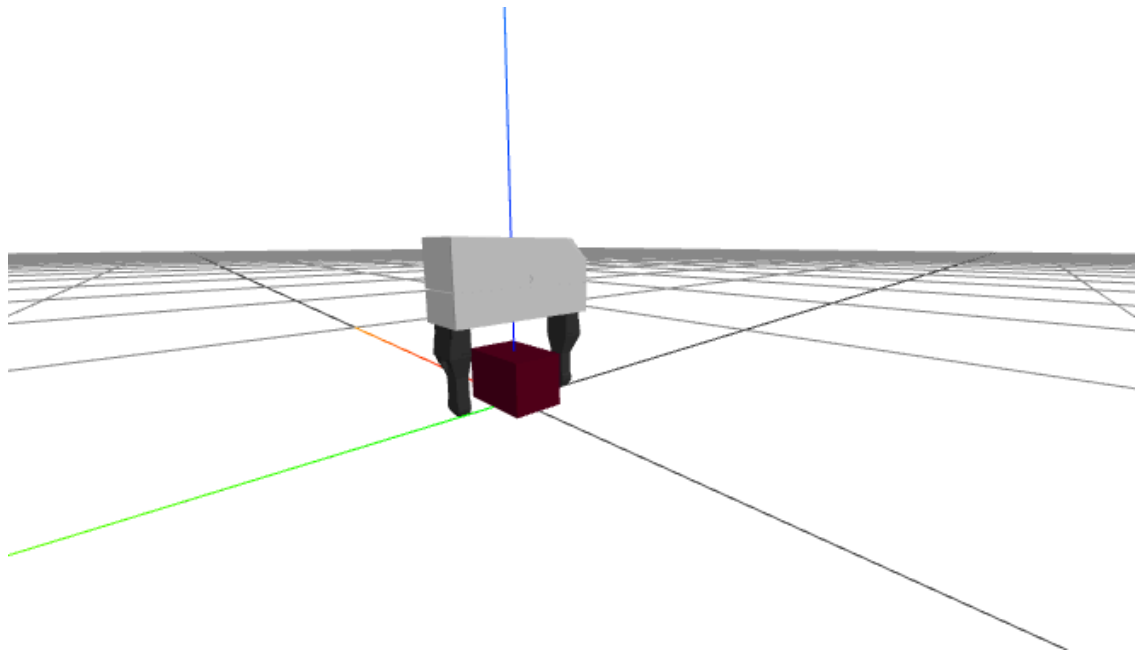
generalized velocity



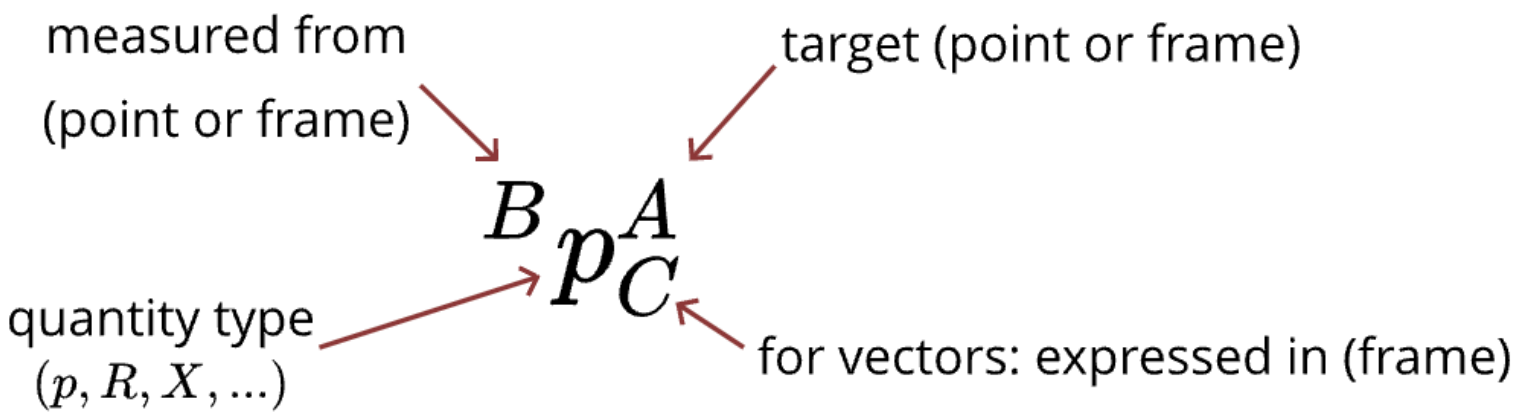
$$v = [J^{G}(q)]^{+W}V^{G}$$

Step 1: Kinematic Frames / Spatial Algebra





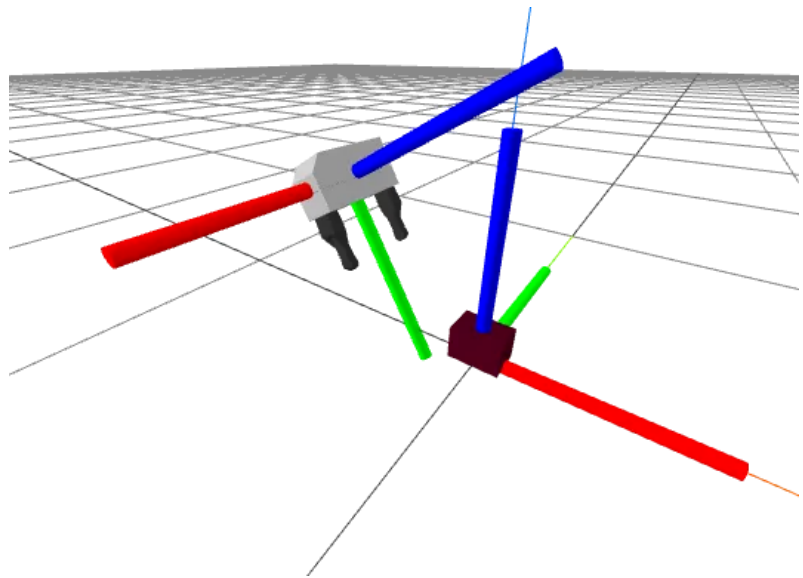
Monogram notation



Check yourself

- G is the gripper frame
- O is the object frame

which is a possible value for ${}^{G}p^{O}$?



a) [.3, 0, -.1]

b) [0, .3, .1]

c) [0, -.3, .1]

Answer: (b)

Check yourself 2

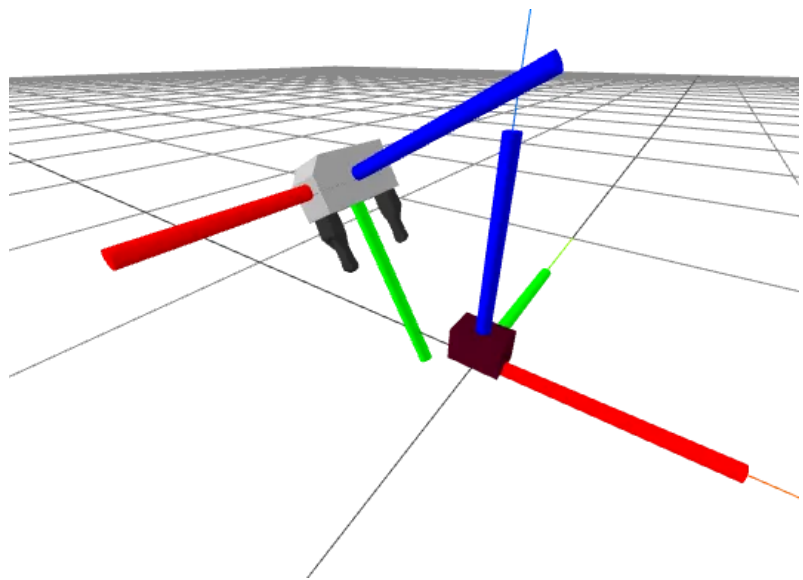
- G is the gripper frame
- O is the object frame

which is a possible value for ${}^{G}p_{O}^{O}$?

a) [.3, 0, -.1]

b) [0, .3, .1]

c) [0, -.3, .1]



Answer: (a)

https://manipulation.mit.edu/pick.html#spatial_algebra

https://manipulation.csail.mit.edu/pick.html#3D_rotation

Representations of Rotation

- Rotation matrix
- Euler angles
- Axis-angle
- Quaternions

Rotations (single axis)

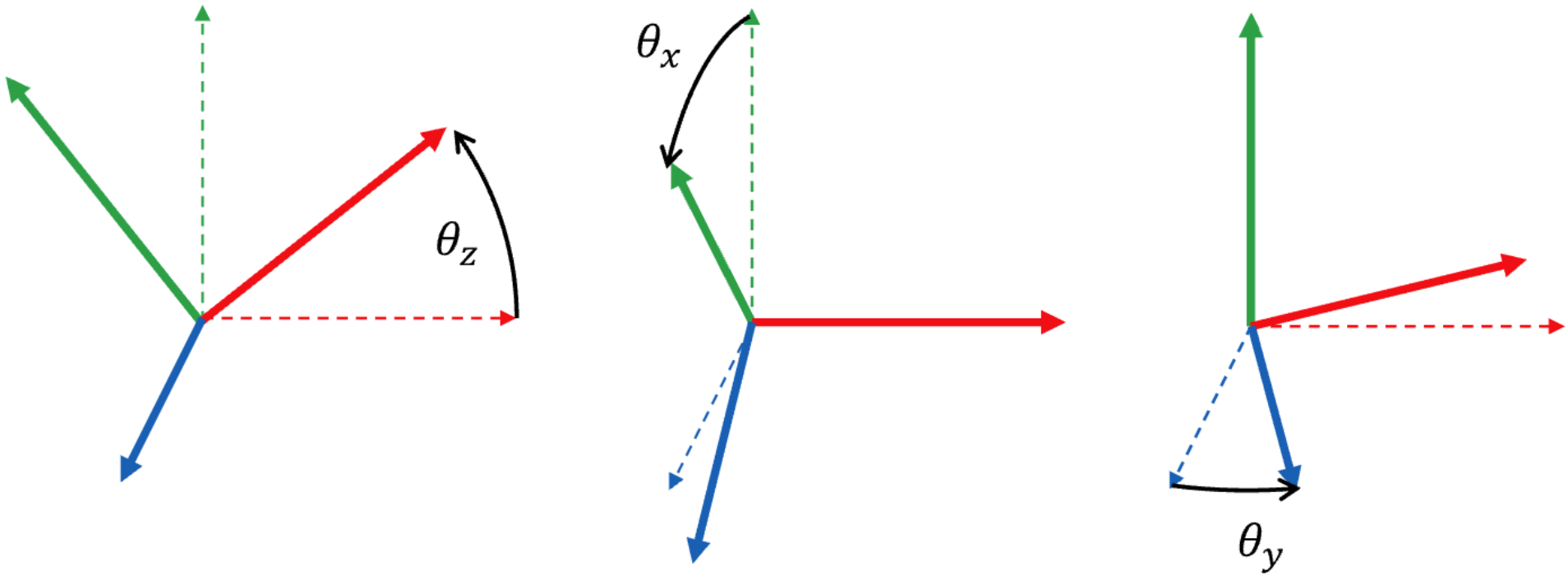
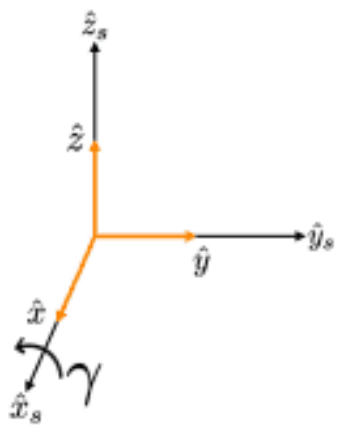
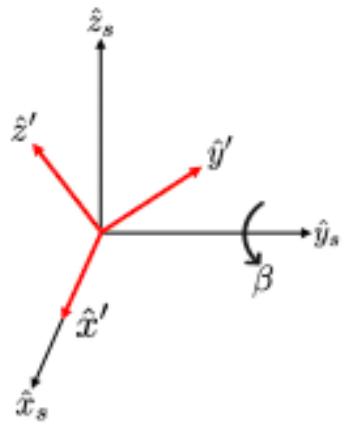


Figure 2. Axis-aligned rotations.

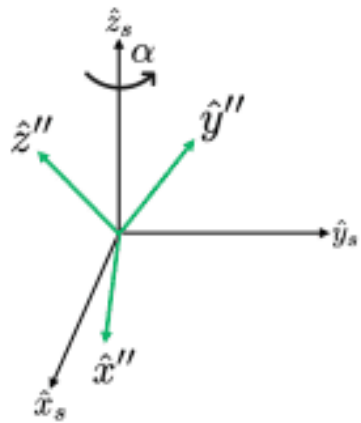
Roll Pitch Yaw (extrinsic)



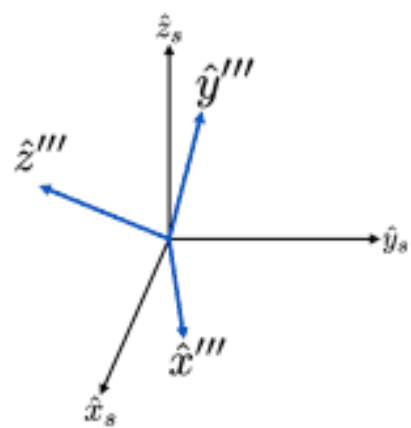
{b} is initially coincident with {s}



frame achieved after a rotation about space frame's x-axis by γ



frame achieved after a rotation about space frame's y-axis by β



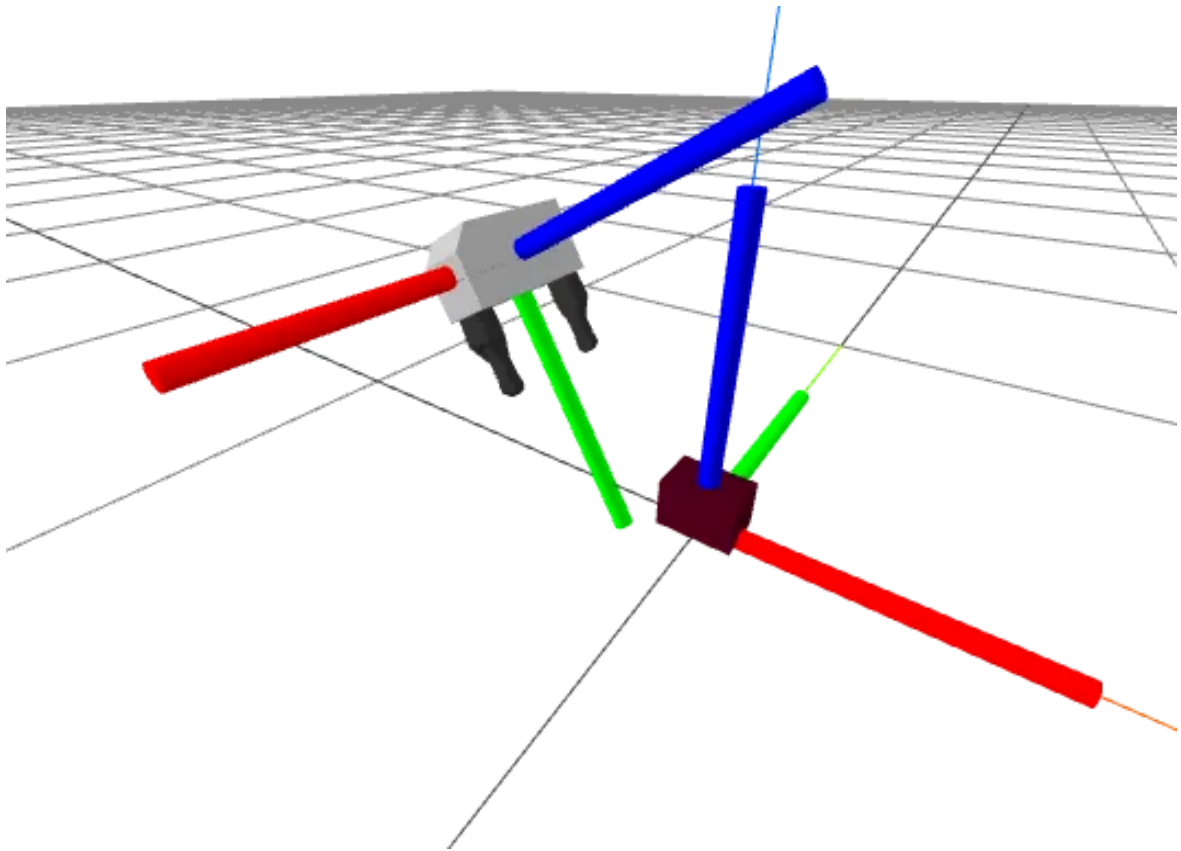
frame achieved after a rotation about space frame's z-axis by α

Mecharithm

Example: Gimbal Lock

Step 2: Gripper Frame Plan "Sketch"

https://manipulation.mit.edu/data/pick_sketch.html



```
def MakeGripperFrames(X_WG, X_WO):
    """
    Takes a partial specification with X_G["initial"] and X_O["initial"] and
    X_O["goal"], and returns a X_G and times with all of the pick and place
    frames populated.
    """
    # Define (again) the gripper pose relative to the object when in grasp.
    p_GgraspO = [0, 0.11, 0]
    R_GgraspO = RotationMatrix.MakeXRotation(
        np.pi / 2.0
    ) @ RotationMatrix.MakeZRotation(np.pi / 2.0)
    X_GgraspO = RigidTransform(R_GgraspO, p_GgraspO)

```

Make sure your trajectory safely clears the bins...

https://manipulation.mit.edu/data/pick_fail.html

https://drake.mit.edu/doxygen_cxx/namespacedrake_1_1trajectories.html

https://drake.mit.edu/doxygen_cxx/classdrake_1_1multibody_1_1_multibody_plant.html#aea0f8aca5f15e90f11e14602b3ba7068